

# Mohammad Raihan Uddin

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## Education

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| <b>Clemson University</b> , PhD, Computer Engineering (Advisor: F. Afghah)                       | Jan 2026 – Present  |
| <b>University of Alabama in Huntsville</b> , MSE, Electrical Engineering (Advisor: D. C. Nguyen) | Jan 2024 – Dec 2025 |
| <b>Daffodil International University</b> , BSc in Electronics & Telecommunication Engineering    | Jan 2013 – Feb 2017 |

## Research Interests

AI-native 5G/6G networks, wireless foundation models, Open RAN security, integrated sensing and communication (ISAC), and trustworthy machine learning for wireless systems.

## Research Experience

**Graduate Research Assistant, IS-WiN Lab, Clemson University** Jan 2026 – Present

- Developed ORAN-DEFEND, a retraining-free subspace detection and sanitization defense against backdoor attacks on deep reinforcement learning (DRL) xApps in Open RAN (accepted, IEEE MECOM 2026).
- Designed RFPrompt, a prompt-based expert adaptation framework for the Large Wireless Model mixture-of-experts backbone, adapting it to modulation classification while training only  $\sim 0.34\%$  of parameters (accepted, IEEE GLOBE-COM 2026).
- Building ISACMoE, a CSI-anchored hierarchical mixture-of-experts model for sensing-assisted communications, and contributing to UniFuse, a fusion framework for wireless foundation models.

**Graduate Research Assistant, NIS Lab, UAH** Jan 2024 – Dec 2025

- Co-developed SimQFL, a quantum federated learning simulator with real-time visualization (IEEE QCE 2025).
- Developed a federated learning framework for detecting false data injection attacks in edge-based smart metering networks (IEEE CCNC 2025).
- Contributed to secure UAV-assisted federated learning using digital twins and zero-knowledge proofs, and authored a comprehensive survey on quantum federated learning (IEEE Communications Surveys & Tutorials).

## Teaching Experience

**Graduate Teaching Assistant, ECE Department, UAH** Jan 2024 – Dec 2025

Led labs and mentored students across seven courses: Computer Organization, Intro to Software Engineering, Intro to Computer Networks, Electric Circuits & Electronics Design Lab, Electric Circuits System, Operating Systems Lab, and Intro to Computer Programming for Engineering Lab.

## Publications

### Journal Articles

1. **M. R. Uddin**, D. C. Nguyen, “Quantum Federated Learning: A Comprehensive Survey,” *IEEE Communications Surveys & Tutorials*.
2. S. Shaon, **M. R. Uddin**, D. C. Nguyen, “Quantum Federated Learning in AI-Native 6G Wireless Networks,” *IEEE Network*.
3. V. K. Quy, S. Shaon, **M. R. Uddin**, N. M. Quy, D. C. Nguyen, “Aerial Intelligent Reflecting Surface (A-IRS) for Wireless Power Transfer in 6G Networks: Security Challenges and Solutions,” *IEEE Network*.
4. M. B. Zami, **M. R. Uddin**, D. C. Nguyen, “Secure UAV-assisted Federated Learning: A Digital Twin-Driven Approach with Zero-Knowledge Proofs,” *IEEE Internet of Things Journal*.
5. **M. R. Uddin**, K. M. Kabir, and M. M. Hasan, “Data Security through Image Processing by Blowfish Algorithm, Genetic Algorithm and LSB,” *International Journal of Computer Applications*, Foundation of Computer Science, vol. 148, 2016.

### Conference Papers

6. R. Rahman, A. Pokharel, **M. R. Uddin**, D. C. Nguyen, “SimQFL: A Quantum Federated Learning Simulator with Real-Time Visualization,” in *Proc. IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2025.
7. **M. R. Uddin**, R. Rahman, D. C. Nguyen, “False Data Injection Attack Detection in Edge-based Smart Metering Networks with Federated Learning,” in *Proc. IEEE Consumer Communications & Networking Conference (CCNC)*, 2025.
8. V. K. Quy, N. M. Quy, T. T. Hoai, S. Shaon, **M. R. Uddin**, D. C. Nguyen, and P. Chatzimisios, “From Federated Learning to Quantum Federated Learning for Space-Air-Ground Integrated Networks,” in *Proc. IEEE Conference on Standards for Communications and Networking (CSCN)*, 2024.

9. M. Arefin, **M. R. Uddin**, N. A. Evan, M. R. Alam, et al., “Enterprise Network: Security Enhancement and Policy Management Using Next-Generation Firewall (NGFW),” in *Computer Networks, Big Data and IoT*, Springer, 2021, pp. 753–769.
10. **M. R. Uddin**, N. A. Evan, M. R. Alam, M. Arefin, et al., “Analysis of Generic Routing Encapsulation (GRE) over IPsec VPN Tunneling in IPv6 Network,” in *International Conference on Ubiquitous Communications and Network Computing*, Springer, 2021, pp. 3–15.
11. S. K. Dey, **M. R. Uddin**, M. Rahman, et al., “Performance Analysis of SDN-based Intrusion Detection Model with Feature Selection Approach,” in *Proc. International Joint Conference on Computational Intelligence*, Springer, 2020, pp. 483–494.
12. N. F. Firoz, M. Arefin, **M. R. Uddin**, et al., “Performance Optimization of Layered Signature Based Intrusion Detection System Using Snort,” in *International Conference on Cyber Security and Computer Science*, Springer, 2020, pp. 14–27.
13. **M. R. Uddin**, K. M. Kabir, and M. T. Arefin, “Artificial Neural Network Inducement for Enhancement of Cloud Computing Security,” in *Proc. 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT)*, IEEE, 2019, pp. 1–6.
14. S. K. Dey, M. M. Rahman, and **M. R. Uddin**, “Detection of Flow Based Anomaly in OpenFlow Controller: Machine Learning Approach in Software Defined Networking,” in *Proc. 4th International Conference on Electrical Engineering and Information & Communication Technology (iCEEiCT)*, IEEE, 2018, pp. 416–421.
15. S. K. Dey, **M. R. Uddin**, K. M. Kabir, and M. M. Rahman, “Enhancing the Security of Cloud Computing: Genetic Algorithm and QR Code Approach,” in *Proc. 4th International Conference on Advances in Electrical Engineering (ICAEE)*, IEEE, 2017, pp. 181–186.

#### Accepted Papers

16. **M. R. Uddin**, T. Seyfi, F. Afghah, “ORAN-DEFEND: Subspace Detection and Sanitization of Backdoor DRL xApps in Open RAN,” accepted to *IEEE MECON 2026*.
17. **M. R. Uddin**, T. Seyfi, F. Afghah, “RFPrompt: Prompt-Based Expert Adaptation of the Large Wireless Model for Modulation Classification,” accepted to *IEEE GLOBECOM 2026*.
18. R. Barker, A. E. Dorcheh, T. Seyfi, **M. R. Uddin**, A. Mohammadhosseini, J. Boone, S. Streit, D. Schlesener, and F. Afghah, “AI-Native Open RAN: A Roadmap from xApps and rApps to Autonomous Network Agents,” accepted to *Allerton Conference 2026*.

#### Under Review

19. **M. R. Uddin**, F. Afghah, “ISACMoE: CSI-Anchored Selective Fusion with Hierarchical Experts for Sensing-Assisted Communications,” submitted to *IEEE ICC 2027*.
20. T. Seyfi, **M. R. Uddin**, F. Afghah, “UniFuse: Unified Wireless Foundation Model Fusion for Channel-Aware Intelligence,” submitted to *ACM MobiCom 2026*.
21. **M. R. Uddin**, D. C. Nguyen, “Next-Generation UAVs: Survey of Emerging Technologies and Security & Privacy,” submitted to *Vehicular Communications*.

#### Awards & Hackathons

**Winner**, SC Quantathon v3, Clemson University (2026); **Participant**, Open6G+AI School & Hackathon, Northeastern University (2026).

#### Technical Skills & Certifications

**Programming & ML:** Python, MATLAB, C++, PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, Qiskit, PennyLane

**Systems & Tools:** Linux, Docker, Kubernetes, AWS, Cisco networking, MySQL, LaTeX

**Certifications:** Certified Kubernetes Administrator (CKA); AWS Certified Solutions Architect – Associate; Docker Certified Associate (DCA); Red Hat Certified Engineer (RHCE); Red Hat Certified System Administrator (RHCSA); Cisco Certified Network Associate (CCNA)

#### Peer Review Service

Reviewed ~60 papers for *IEEE IoT Journal*, *ACM Computing Surveys*, *IEEE TDSC*, *IEEE TNSE*, *IEEE TCCN*, *IEEE TGCN*, *IEEE OJ-COMS*, *IEEE Communications Letters*, *IEEE Network*, *IEEE IoT Magazine*, *IEEE Vehicular Technology Magazine*, *Computer Networks*, and conferences *IEEE/CVF WACV 2027* and *IEEE INFOCOM NetRobiCS Workshop 2026*.

#### Leadership & Service

Head Organizer, TEDxUAHuntsville 2025 • President, Bangladeshi Student Association, UAH (2024–2026) • Secretary, Graduate Student Association, UAH (2024–2025) • President (2022–2023) and General Secretary (2019–2021), House of Youth Dialogue • IEEE Member (2015–Present)